

A Non-Ergodic Good Sequence With Purely Imaginary Nontrivial Spectrum

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Source and Question

The source paper is Cuny, Eisner, and Farkas, *Wiener's lemma along primes and other subsequences*, arXiv:1701.00101. In Definition 2.2, a subsequence (k_n) of $\mathbb{N}$ is called *good* if

$$c(\lambda) := \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \lambda^{k_n}$$

exists for every $\lambda \in \mathbb{T}$. It is *ergodic* if $c = \mathbf{1}_{\{1\}}$.

Question 2.7 asks two questions. The second is:

Is there a non-ergodic, good sequence (k_n) with $\Re c(\lambda) = 0$ for each $\lambda \in \mathbb{T} \setminus \{1\}$?

The following questions arise naturally, cf. also Proposition 2.13 below.

Question 2.7. Does the existence of $\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N |\widehat{\mu}(k_n)|^2$ for every probability Borel measure μ imply that (k_n) is good? Is there a non-ergodic, good sequence (k_n) with $\Re c(\lambda) = 0$ for each $\lambda \in \mathbb{T} \setminus \{1\}$?

Remark 2.8. If one replaces “probability measure” by “complex measure”, the answer to the first question is positive. Indeed, it is easy to see that for a subsequence (k_n) of $\mathbb{N}$ the following assertions are equivalent.

Result

Theorem 1. *There exists a non-ergodic good subsequence (k_n) of $\mathbb{N}$ such that*

$$\Re c(\lambda) = 0 \quad \forall \lambda \in \mathbb{T} \setminus \{1\}.$$

Thus the second question in Question 2.7 has an affirmative answer.

Proof Intuition

It is enough to build a set of integers whose Fourier-Bohr coefficients are supported, away from 1, only at two conjugate roots of unity, and whose two coefficients are purely imaginary. A deterministic periodic residue distribution modulo 3 gives exactly the desired coefficients at the cubic roots, but a purely periodic set would introduce unwanted coefficients at other roots of unity. Randomizing one residue class removes these extra frequencies: the centered random part has uniformly vanishing exponential sums almost surely, while the periodic expectation keeps the desired cubic-root coefficients.

Proof

Let $\omega = \exp(2\pi i/3)$. Define a random subset $A \subset \mathbb{N}$ as follows, independently for different integers:

$$\mathbb{P}(m \in A) = \begin{cases} 1/2, & m \equiv 0 \pmod{3}, \\ 1, & m \equiv 1 \pmod{3}, \\ 0, & m \equiv 2 \pmod{3}. \end{cases}$$

Let $1_A(m)$ be its indicator and put

$$d_m = \begin{cases} 1/2, & m \equiv 0 \pmod{3}, \\ 1, & m \equiv 1 \pmod{3}, \\ 0, & m \equiv 2 \pmod{3}. \end{cases}$$

The deterministic mean density is

$$\delta = \frac{1}{3} \left(\frac{1}{2} + 1 + 0 \right) = \frac{1}{2}.$$

Lemma 1 (Uniform cancellation of the centered random part). *Almost surely,*

$$\sup_{\lambda \in \mathbb{T}} \left| \frac{1}{M} \sum_{m=1}^M (1_A(m) - d_m) \lambda^m \right| \rightarrow 0.$$

Proof. Write $X_m = 1_A(m) - d_m$. These are independent, mean-zero, real, and bounded by 1 in absolute value. For a fixed $\lambda \in \mathbb{T}$, Hoeffding's inequality applied to real and imaginary parts gives

$$\mathbb{P} \left(\left| \sum_{m=1}^M X_m \lambda^m \right| > \varepsilon M \right) \leq 4 \exp(-c\varepsilon^2 M)$$

with an absolute constant $c > 0$.

Let $\mathcal{G}_M$ be a grid on $\mathbb{T}$ with mesh at most ε/M and cardinality $O(M/\varepsilon)$. By the union bound,

$$\mathbb{P} \left(\max_{\zeta \in \mathcal{G}_M} \left| \sum_{m=1}^M X_m \zeta^m \right| > \varepsilon M \right) \leq C_\varepsilon M \exp(-c\varepsilon^2 M),$$

which is summable in M . Borel–Cantelli implies that, almost surely, the grid maximum is at most εM for all large M .

If λ is within ε/M in angular distance of $\zeta \in \mathcal{G}_M$, then

$$\left| \sum_{m=1}^M X_m (\lambda^m - \zeta^m) \right| \leq \sum_{m=1}^M m |\lambda - \zeta| \leq C\varepsilon M.$$

Since $\varepsilon > 0$ is arbitrary (take a countable sequence $\varepsilon \downarrow 0$), the asserted uniform convergence follows. $\square$

Fix a realization of A for which the lemma holds. Then

$$\frac{1}{M} \sum_{m=1}^M 1_A(m) \lambda^m - \frac{1}{M} \sum_{m=1}^M d_m \lambda^m \longrightarrow 0$$

uniformly in λ .

The deterministic periodic average is immediate. If $\lambda^3 \neq 1$, then

$$\frac{1}{M} \sum_{m=1}^M d_m \lambda^m \longrightarrow 0.$$

If $\lambda^3 = 1$, then

$$\frac{1}{M} \sum_{m=1}^M d_m \lambda^m \longrightarrow \frac{1}{3} \left(\frac{1}{2} + \lambda \right).$$

In particular, at $\lambda = 1$ this limit is $\delta = 1/2$, so A has natural density $1/2$.

Let (k_n) be the increasing enumeration of A . Since A has positive natural density, for every $\lambda \in \mathbb{T}$,

$$\frac{1}{N} \sum_{n=1}^N \lambda^{k_n} = \frac{1}{|A \cap [1, k_N]|} \sum_{\substack{m \leq k_N \\ m \in A}} \lambda^m \longrightarrow c(\lambda),$$

where

$$c(\lambda) = \begin{cases} 1, & \lambda = 1, \\ 0, & \lambda^3 \neq 1, \\ \frac{1}{3\delta} \left(\frac{1}{2} + \lambda \right), & \lambda^3 = 1. \end{cases}$$

For the two nontrivial cubic roots this gives

$$c(\omega) = \frac{2}{3} \left(\frac{1}{2} + \omega \right) = \frac{i\sqrt{3}}{3}, \quad c(\omega^2) = \frac{2}{3} \left(\frac{1}{2} + \omega^2 \right) = -\frac{i\sqrt{3}}{3}.$$

Thus (k_n) is good and not ergodic, because $c(\omega) \neq 0$. Moreover, the displayed formula shows that every nontrivial coefficient is either 0 or purely imaginary. Hence $\Re c(\lambda) = 0$ for every $\lambda \neq 1$, as required.

Verification Notes

The construction is non-explicit only in the harmless probabilistic-method sense: the proof shows that almost every random realization works, so at least one deterministic increasing subsequence exists. The script `code/verify_residue_coefficients.py` verifies the finite residue calculation

$$c(1) = 1, \quad c(\omega) = i\sqrt{3}/3, \quad c(\omega^2) = -i\sqrt{3}/3.$$

The first sentence of Question 2.7, about whether the existence of the quadratic averages for all probability measures implies goodness, is not addressed here.

Novelty Check

The lightweight run indexes were searched for arXiv id 1701.00101, the title, `Wiener's lemma`, `good sequence`, `non-ergodic`, and `Rec(lambda)`. Web searches for exact phrases from Question 2.7 and the source title found no later explicit answer in the bounded search performed on 2026-07-03. This packet should be reviewed as a new full answer to the second part of the source question.

References

References

- [1] C. Cuny, T. Eisner, and B. Farkas. Wiener's lemma along primes and other subsequences. arXiv:1701.00101, 2017.
- [2] W. Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963.